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(54) **SPEECH ENCODERS WITH THINK TOKENS**

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(52) **U.S. Cl.**

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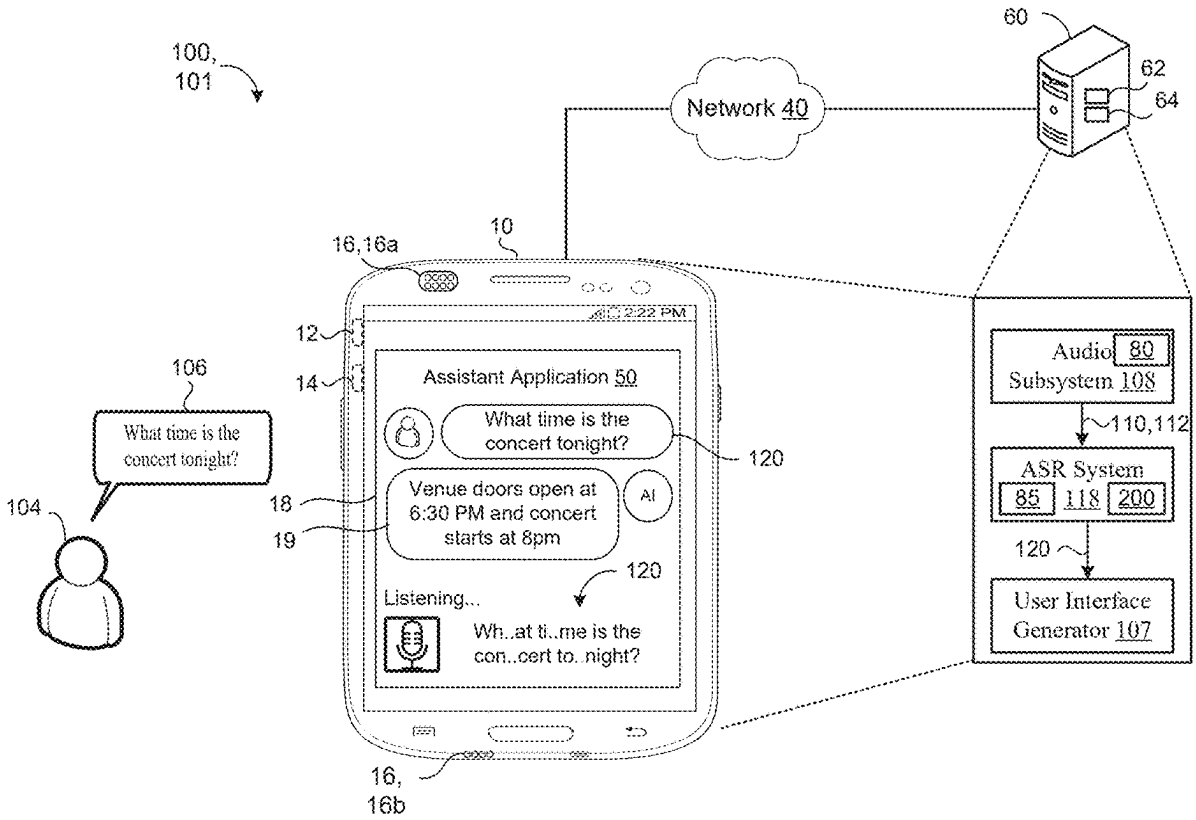
(57) **ABSTRACT**

A method includes receiving a sequence of input tokens and inserting a sequence of special think tokens into the sequence of input tokens. The inserted special tink tokens are interleaved with the sequence of input tokens. The method also includes processing, using an encoder, the sequence of input tokens interleaved with the sequence of special think tokens to generate a sequence of encoder features. The method also includes processing the sequence of encoder features to generate an output result.



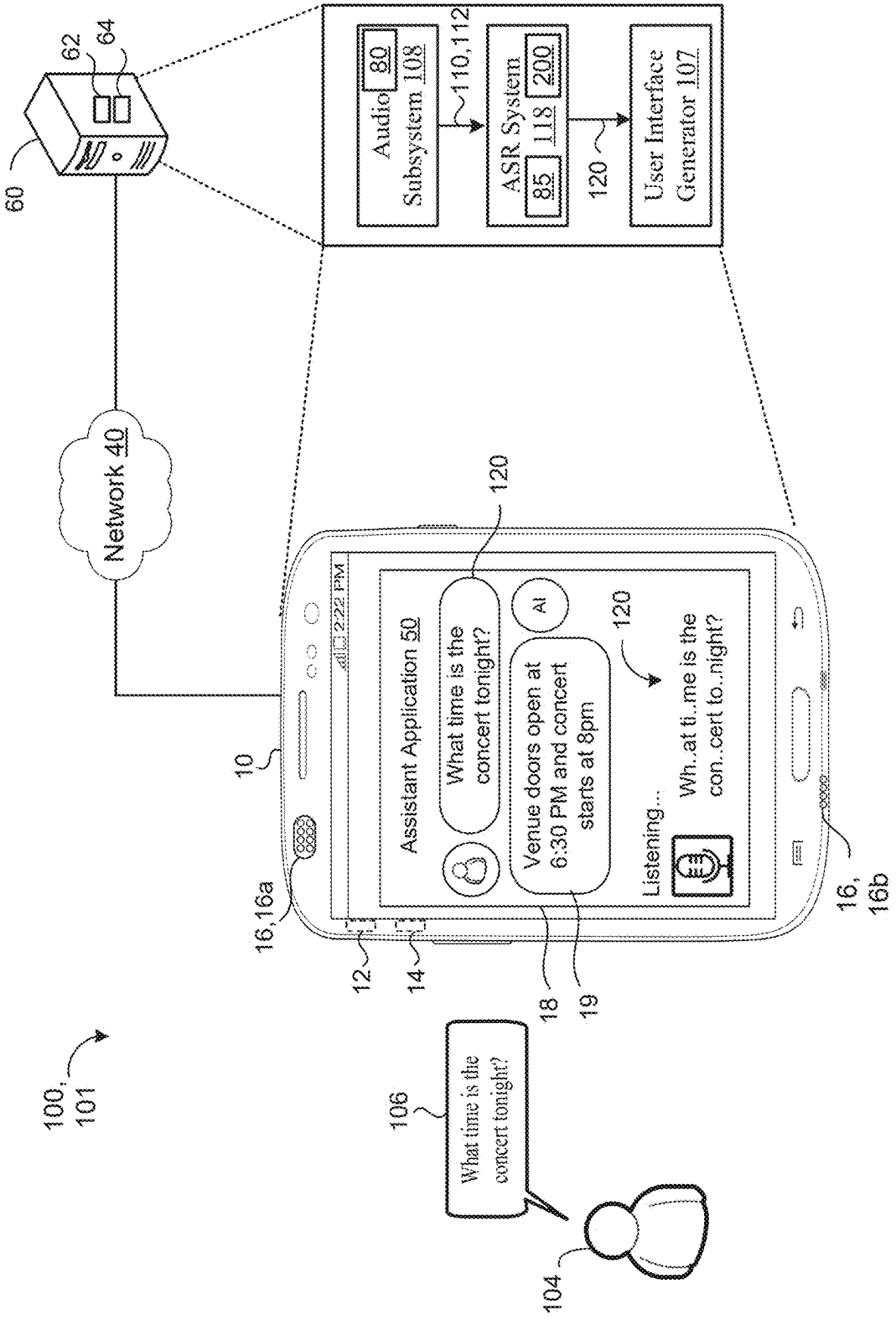

100, 101
60
62
64
Network 40
110,112
Audio Subsystem 108
80
ASR System
85
118
200
120
User Interface Generator 107
10
16,16a
16, 16b
12
14
18
19
2:22 PM
Assistant Application 50
What time is the concert tonight?
Venue doors open at 6:30 PM and concert starts at 8pm
AI
Listening...
Wh..at ti..me is the con..cert to..night?
106
What time is the concert tonight?
104

FIG. 1

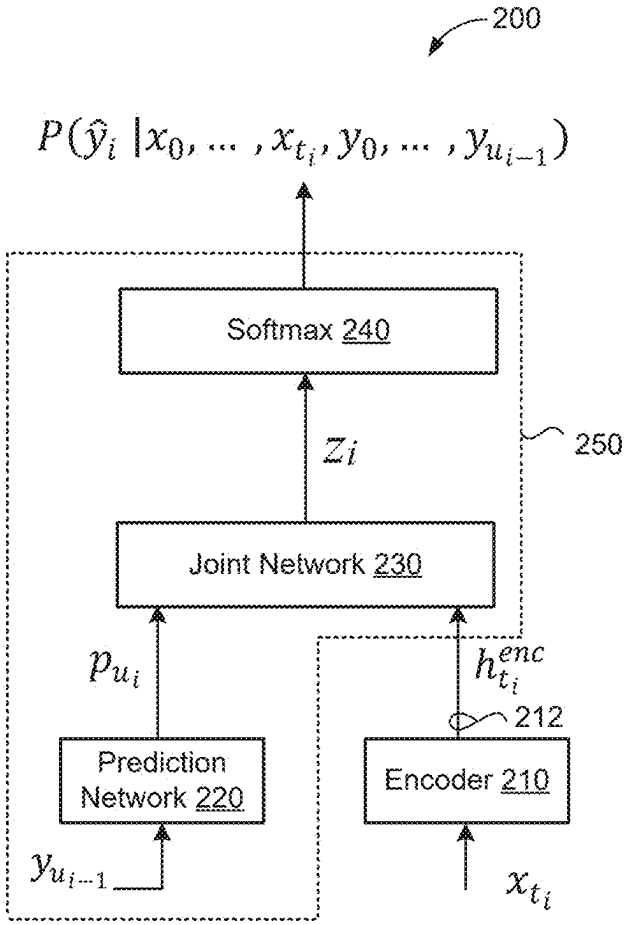

200
$P(\hat{y}_i | x_0, \ldots, x_{t_i}, y_0, \ldots, y_{u_{i-1}})$
Softmax 240
z_i
250
Joint Network 230
p_{u_i}
$h_{t_i}^{enc}$
212
Prediction Network 220
Encoder 210
$y_{u_{i-1}}$
x_{t_i}

FIG. 2

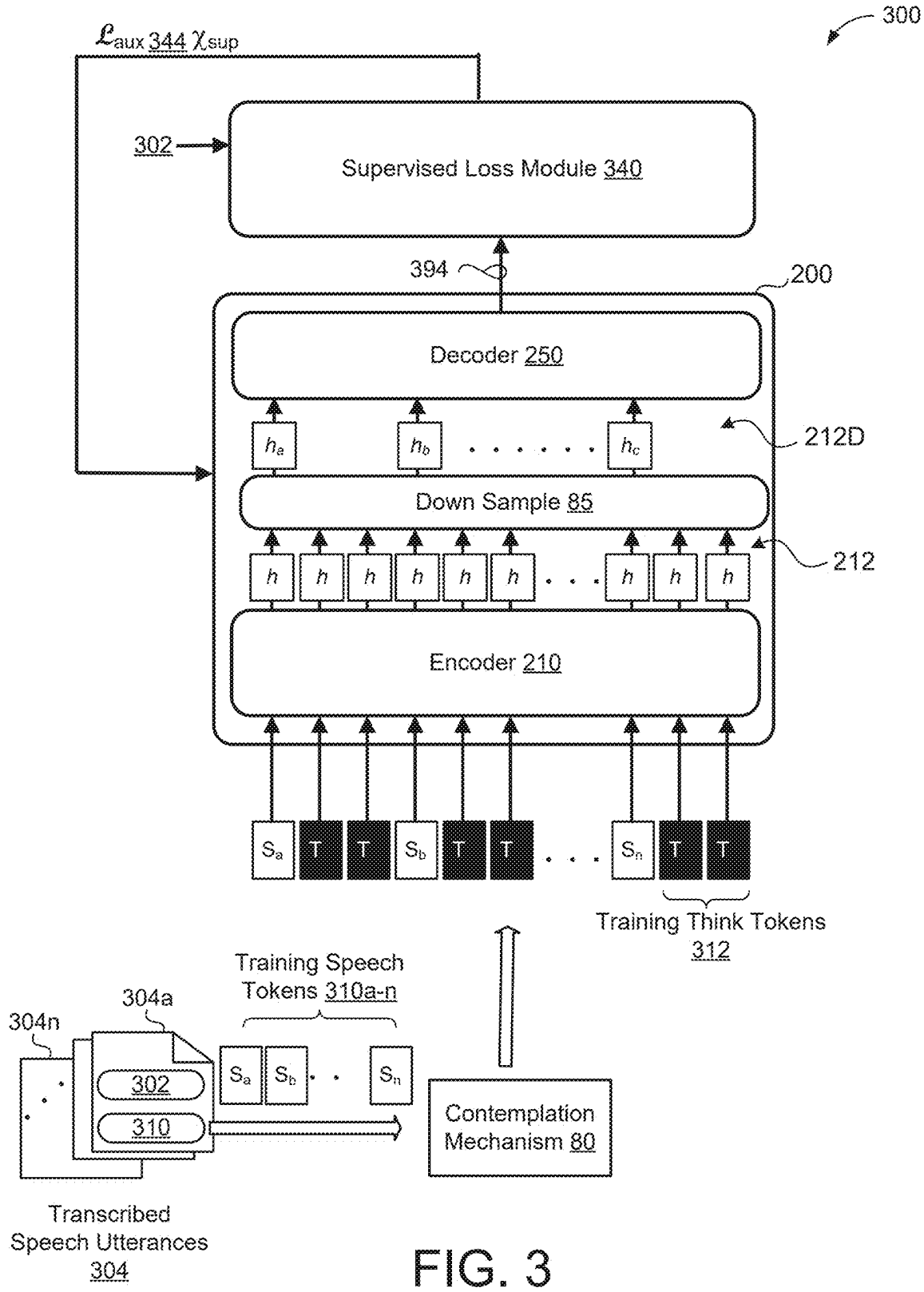

300
$\mathcal{L}_{aux}$ 344 χ_{sup}
302
Supervised Loss Module 340
394
200
Decoder 250
h_a
h_b
h_c
212D
Down Sample 85
h
212
Encoder 210
S_a
T
S_b
S_n
Training Think Tokens 312
Training Speech Tokens 310a-n
304a
304n
302
310
Contemplation Mechanism 80
Transcribed Speech Utterances 304

FIG. 3

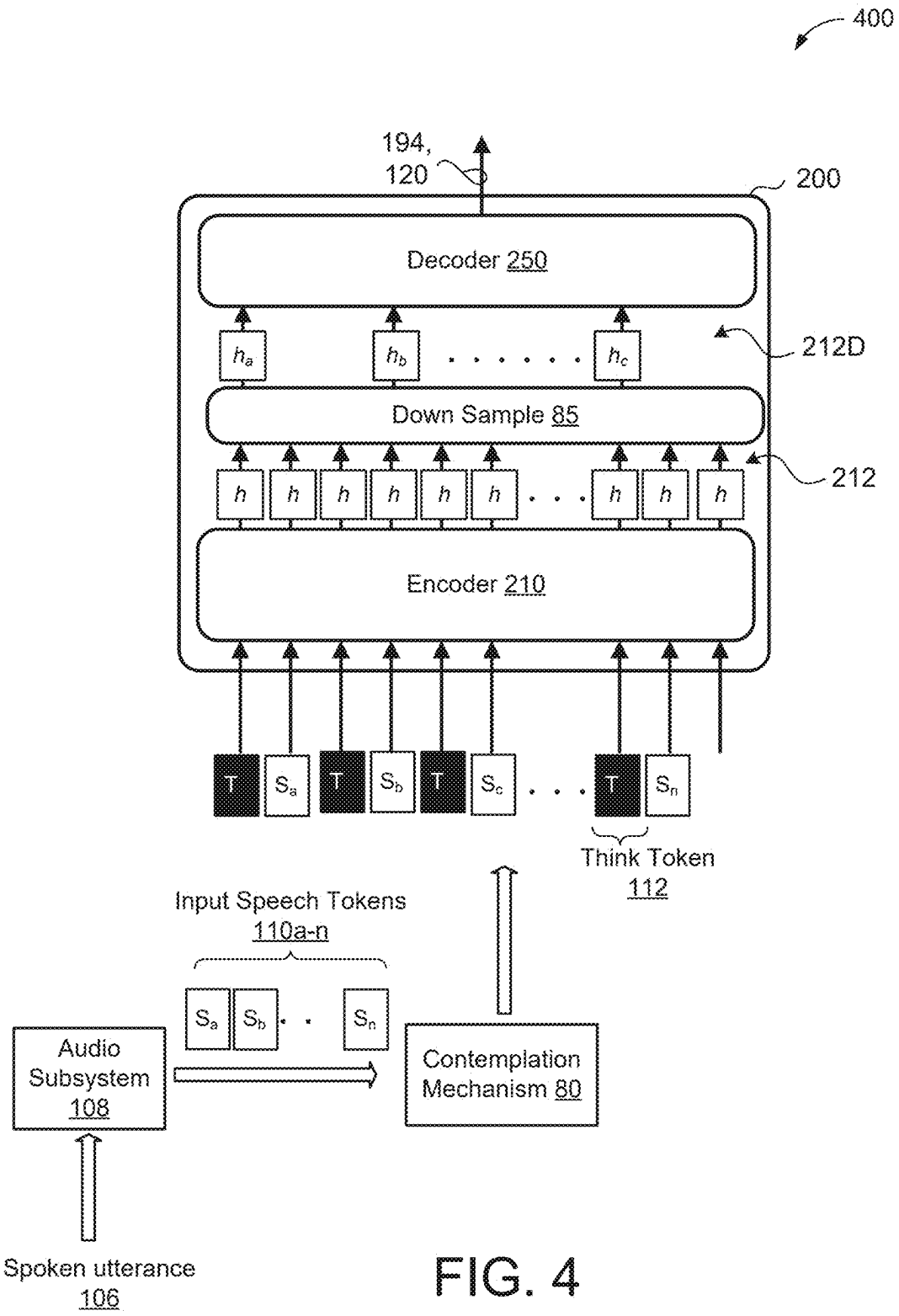

400
194, 120
200
Decoder 250
h_a
h_b
h_c
212D
Down Sample 85
h
212
Encoder 210
T
S_a
S_b
S_c
S_n
Think Token 112
Input Speech Tokens 110a-n
Audio Subsystem 108
Contemplation Mechanism 80
Spoken utterance 106

FIG. 4

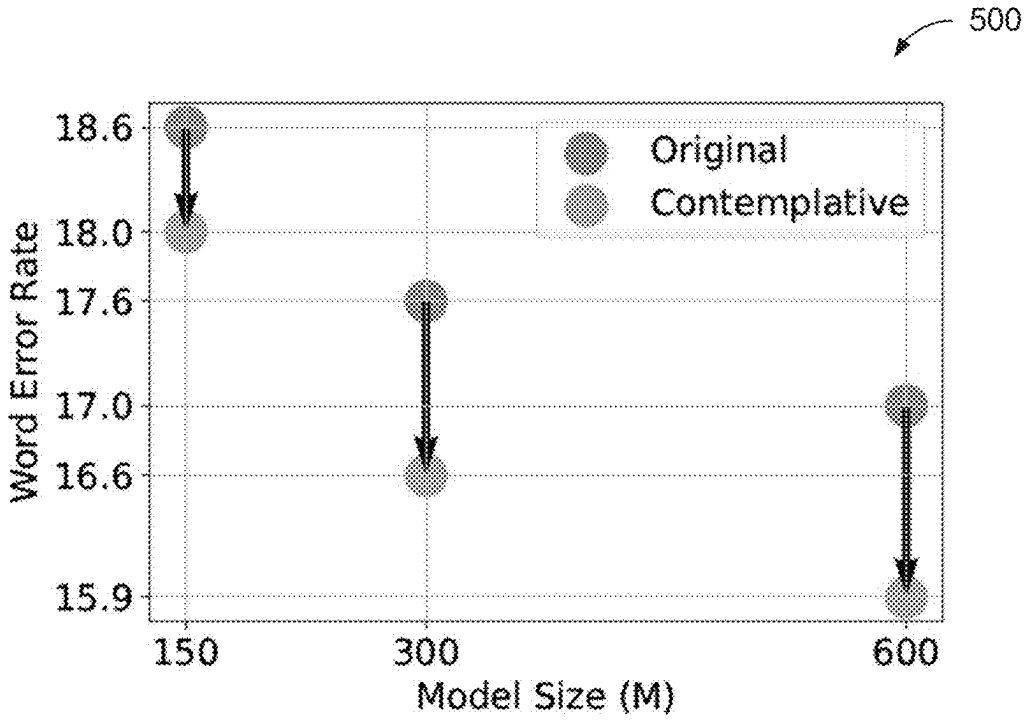

500
18.6
18.0
17.6
17.0
16.6
15.9
Word Error Rate
Original
Contemplative
150
300
600
Model Size (M)

FIG. 5

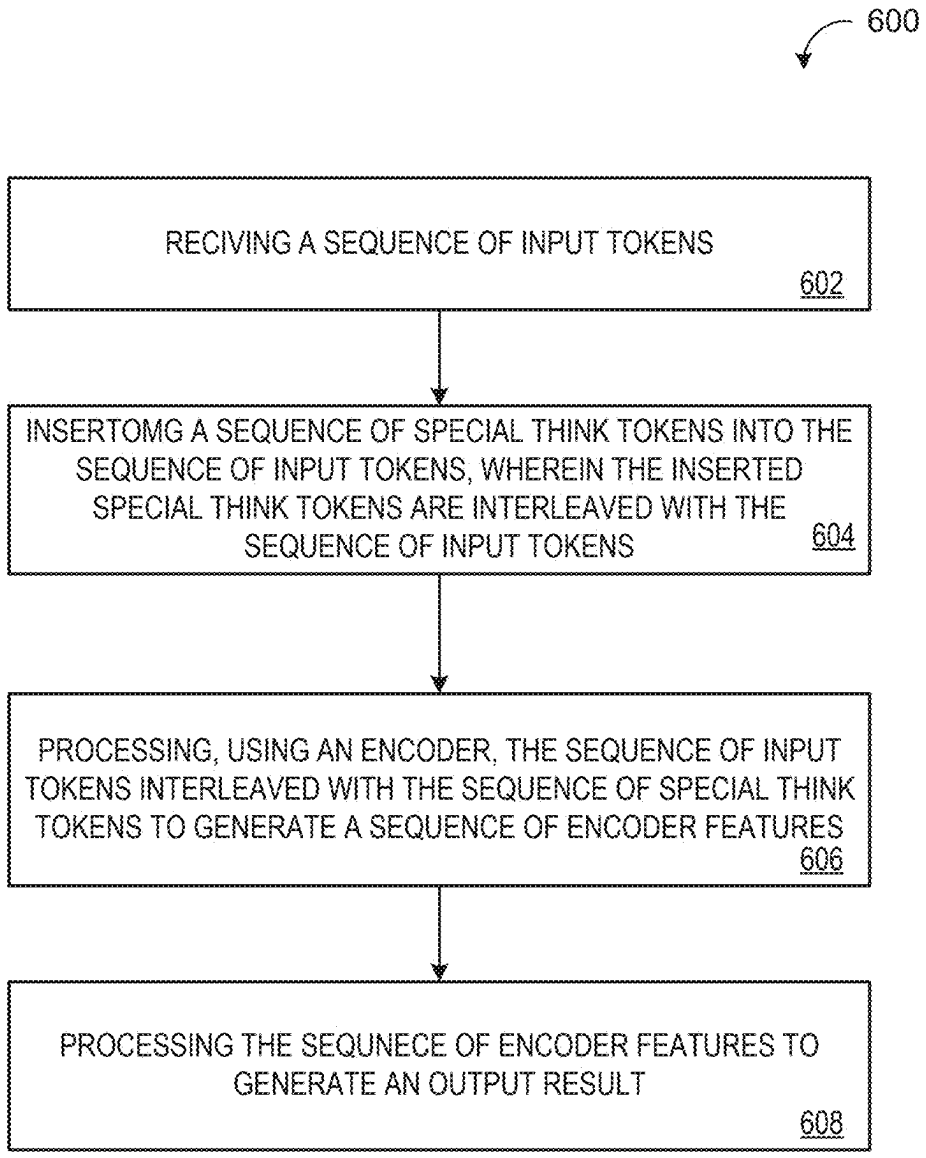

600
RECIVING A SEQUENCE OF INPUT TOKENS
602
INSERTOMG A SEQUENCE OF SPECIAL THINK TOKENS INTO THE SEQUENCE OF INPUT TOKENS, WHEREIN THE INSERTED SPECIAL THINK TOKENS ARE INTERLEAVED WITH THE SEQUENCE OF INPUT TOKENS
604
PROCESSING, USING AN ENCODER, THE SEQUENCE OF INPUT TOKENS INTERLEAVED WITH THE SEQUENCE OF SPECIAL THINK TOKENS TO GENERATE A SEQUENCE OF ENCODER FEATURES
606
PROCESSING THE SEQUNECE OF ENCODER FEATURES TO GENERATE AN OUTPUT RESULT
608

FIG. 6

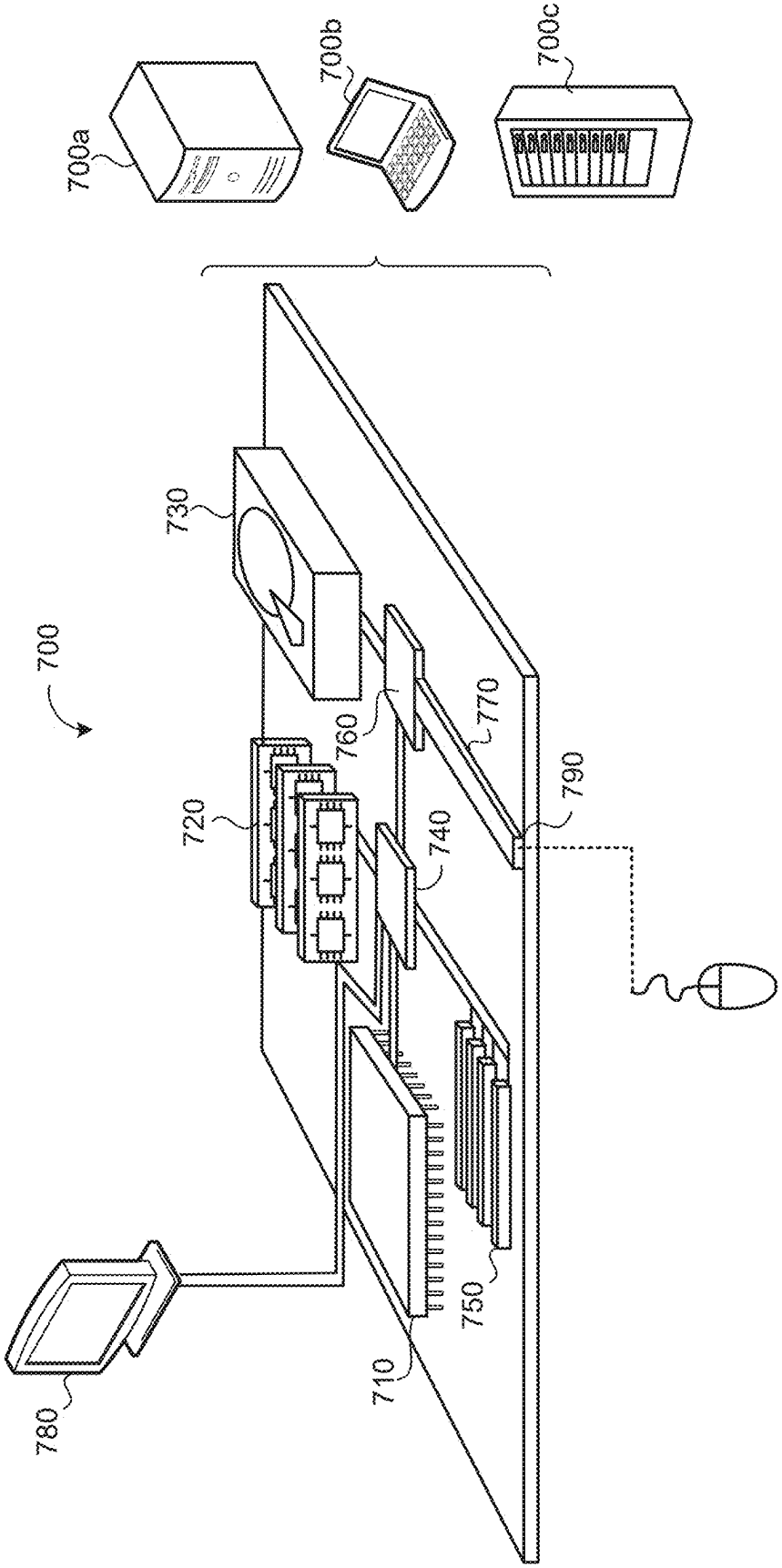


FIG. 7

SPEECH ENCODERS WITH THINK TOKENS

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This U.S. patent application claims priority under 35 U.S.C. § 119 (e) to U.S. Provisional Application 63/559,811, filed on Feb. 29, 2024. The disclosure of this prior application is considered part of the disclosure of this application and is hereby incorporated by reference in its entirety.

TECHNICAL FIELD

[0002] This disclosure relates to speech encoders with think tokens.

BACKGROUND

[0003] End-to-end automatic speech recognition (ASR) models have seen revolutionary quality gains with the recent development of large-scale universal speech models (USM). Encoders play a pivotal role in ASR models, serving as the gateway for feature extraction and representation. Notably, increasing per-token computation of these extracted features within the encoder is known to enhance feature representation quality. While conventional techniques primarily aim to increase the size of the encoder to increase per token computation, these techniques dramatically increase the parameter count of the encoder model, and thus, especially pose challenges of large-scale USMs which are already at massive scale with parameter counts capable of exceeding several billion parameters.

SUMMARY

[0004] One aspect of the disclosure provides a computer-implemented method that when executed on data processing hardware causes the data processing hardware to perform operations that include receiving a sequence of input tokens and inserting a sequence of special think tokens into the sequence of input tokens, wherein the inserted special think tokens are interleaved with the sequence of input tokens. The operations also include processing, using an encoder, the sequence of input tokens interleaved with the sequence of special think tokens to generate a sequence of encoder features and processing the sequence of encoder features to generate an output result.

[0005] Implementations of the disclosure may include one or more of the following optional features. In some implementations, inserting the sequence of special tokens includes, for each corresponding input token in the sequence of input tokens, prepending a fixed number of the special think tokens to the corresponding input token. In other implementations, inserting the sequence of special tokens includes, for each corresponding input token in the sequence of input tokens, prepending a fixed number of the special think tokens to the corresponding input token.

[0006] In some examples, the sequence of input tokens comprises a sequence of speech tokens characterizing a spoken utterance and the encoder includes a speech encoder. Here, the output result may include a transcription of the spoken utterance or a synthetic speech translation of the spoken utterance.

[0007] The special think tokens may include handcrafted think tokens. Alternatively, the sequence of special think tokens include duplications of the sequence of input tokens, wherein inserting the sequence of special tokens includes, for each corresponding input token in the sequence of input tokens: duplicating the corresponding input token; and appending or prepending one or more duplications of the corresponding input token to the corresponding input token. Optionally, the special think tokens may include learnable parameters.

[0008] In some implementations, a number of the encoder features in the sequence of encoder features is equal to a sum of a number of the input tokens in the sequence of input tokens and a number of the special think tokens in the sequence of special think tokens. In these implementations, processing the sequence of encoder features to generate the output result includes down-sampling the sequence of encoder features to obtain a down-sampled encoder feature sequence and processing, using a decoder, the down-sampled encoder feature sequence to generate the output result. Here, a number of the encoder features in the down-sampled encoder feature sequence is equal to the number of the input tokens in the sequence of input tokens.

[0009] In some examples, processing the sequence of encoder features to generate the output result includes processing, using a decoder, the sequence of encoder features to generate the output result. Here, a number of the encoder features in the sequence of encoder features may be greater than a number of the input tokens in the sequence of input tokens.

[0010] A model implementing the encoder may be trained by a training process that includes: obtaining a set of set of training input token sequences, each training input token sequence including a corresponding sequence of training input tokens; for each training input token sequence: inserting a corresponding sequence of training think tokens into the training input token sequence such that corresponding sequence of training think tokens are interleaved with the corresponding sequence of training input tokens; and processing, using the encoder, the training input token sequence interleaved with the corresponding sequence of training think tokens to generate a sequence of training encoder features; and training the model based on the sequence of training encoder features generated for each training input token sequence. A ratio of the corresponding sequence of training think tokens inserted into each training input token sequence gradually increases during the training process. The model may include a recurrent neural network-transducer (RNN-T) architecture including the encoder, a prediction network, and a joint network.

[0011] Another aspect of the disclosure provides a system that includes data processing hardware and memory hardware storing instructions that when executed on the data processing hardware causes the data processing hardware to perform operations that include receiving a sequence of input tokens and inserting a sequence of special think tokens into the sequence of input tokens, wherein the inserted special think tokens are interleaved with the sequence of input tokens. The operations also include processing, using an encoder, the sequence of input tokens interleaved with the sequence of special think tokens to generate a sequence of encoder features and processing the sequence of encoder features to generate an output result.

[0012] This aspect of the disclosure may include one or more of the following optional features. In some implementations, inserting the sequence of special tokens includes, for each corresponding input token in the sequence of input

tokens, prepending a fixed number of the special think tokens to the corresponding input token. In other implementations, inserting the sequence of special tokens includes, for each corresponding input token in the sequence of input tokens, prepending a fixed number of the special think tokens to the corresponding input token.

[0013] In some examples, the sequence of input tokens comprises a sequence of speech tokens characterizing a spoken utterance and the encoder includes a speech encoder Here, the output result may include a transcription of the spoken utterance or a synthetic speech translation of the spoken utterance.

[0014] The special think tokens may include handcrafted think tokens. Alternatively, the sequence of special think tokens include duplications of the sequence of input tokens, wherein inserting the sequence of special tokens includes, for each corresponding input token in the sequence of input tokens: duplicating the corresponding input token; and appending or prepending one or more duplications of the corresponding input token to the corresponding input token. Optionally, the special think tokens may include learnable parameters.

[0015] In some implementations, a number of the encoder features in the sequence of encoder features is equal to a sum of a number of the input tokens in the sequence of input tokens and a number of the special think tokens in the sequence of special think tokens. In these implementations, processing the sequence of encoder features to generate the output result includes down-sampling the sequence of encoder features to obtain a down-sampled encoder feature sequence and processing, using a decoder, the down-sampled encoder feature sequence to generate the output result. Here, a number of the encoder features in the down-sampled encoder feature sequence is equal to the number of the input tokens in the sequence of input tokens.

[0016] In some examples, processing the sequence of encoder features to generate the output result includes processing, using a decoder, the sequence of encoder features to generate the output result. Here, a number of the encoder features in the sequence of encoder features may be greater than a number of the input tokens in the sequence of input tokens.

[0017] A model implementing the encoder may be trained by a training process that includes: obtaining a set of set of training input token sequences, each training input token sequence including a corresponding sequence of training input tokens, for each training input token sequence: inserting a corresponding sequence of training think tokens into the training input token sequence such that corresponding sequence of training think tokens are interleaved with the corresponding sequence of training input tokens; and processing, using the encoder, the training input token sequence interleaved with the corresponding sequence of training think tokens to generate a sequence of training encoder features; and training the model based on the sequence of training encoder features generated for each training input token sequence. A ratio of the corresponding sequence of training think tokens inserted into each training input token sequence gradually increases during the training process. The model may include a recurrent neural network-transducer (RNN-T) architecture including the encoder, a prediction network, and a joint network.

[0018] The details of one or more implementations of the disclosure are set forth in the accompanying drawings and the description below. Other aspects, features, and advantages will be apparent from the description and drawings, and from the claims.

DESCRIPTION OF DRAWINGS

[0019] FIG. **1** is a schematic view of an example system for performing speech recognition.

[0020] FIG. **2** is a schematic view of a Recurrent Neural Network-Transducer (RNN-T) model architecture.

[0021] FIG. **3** is an example training process for training an audio encoder of a speech recognition model to increase a per speech token computation of the audio encoder.

[0022] FIG. **4** is a schematic view of a speech recognition model utilizing a contemplative mechanism to insert special think tokens into a sequence of speech tokens to increase per speech token computation by an audio encoder.

[0023] FIG. **5** is an example plot depicting speech recognition performance of models of varying size with and without utilizing a contemplative mechanism.

[0024] FIG. **6** is a flowchart of an exemplary arrangement of operations for a method of increasing computation per speech token within an encoder while maintaining an original parameter count of the encoder.

[0025] FIG. **7** is a schematic view of an example computing device that may be used to implement the systems and methods described herein.

[0026] Like reference symbols in the various drawings indicate like elements.

DETAILED DESCRIPTION

[0027] End-to-end automatic speech recognition (ASR) models have seen revolutionary quality gains with the recent development of large-scale universal speech models (USM). Encoders play a pivotal role in ASR models, serving as the gateway for feature extraction and representation. When the encoder is an audio encoder, the extracted features may be referred to as 'speech tokens" each corresponding to a respective audio feature (e.g., fixed-length audio frame) in a sequence of extracted audio features characterizing a spoken utterance.

[0028] Notably, increasing per-token computation of these extracted features within the encoder is known to enhance feature representation quality. While conventional techniques primarily aim to increase the size of the encoder to increase per token computation, these techniques dramatically increase the parameter count of the encoder model, and thus, especially pose challenges of large-scale USMs which are already at massive scale with parameter counts capable of exceeding several billion parameters. That is, further increasing the parameter count of large-scale USMs introduces complexities beyond mere computational demand such as hindering storage, data transmission, and efficient loading of data into virtual random access memory (VRAM) for execution by underlying data processing hardware, e.g., graphical processing units (GPUs) and/or tensor processing units (TPUs).

[0029] Implementations herein are directed toward a strategic technique of increasing computation per speech token within an encoder while maintaining an original parameter count of the encoder. Specifically, implementations are directed toward receiving a sequence of input tokens to be encoded by the encoder and inserting a sequence of special think tokens into the sequence of input tokens such that the

inserted special think tokens are interleaved within the sequence of input tokens, and processing, using the encoder, the sequence of input tokens interleaved with the sequence of special think tokens to generate a sequence of encoder features. Thereafter, further processing is performed on the sequence of encoder features to generate an output result. The sequence of special tokens may be inserted by either appending or prepending a fixed number of the special think tokens to each corresponding input token. As a result, a number of the encoder features in the sequence of encoder features is equal to a sum of a number of the input tokens in the sequence of input tokens and an umber of the special tokens in the sequence of special tokens. Since the number of encoder features is greater than the number of the input tokens in the sequence of input tokens, the processing of the sequence of input features to generate the output result may include down-sampling the sequence of the encoder features to obtain a down-sampled encoder feature sequence having a same number of encoder features as the number of the input tokens in the sequence of input tokens and processing, using a decoder, the down-sampled encoder feature sequence to generate the output result.

[0030] As will become apparent, the interleaving of the special think tokens within the sequence of input tokens increases per token computation by the encoder to unlock performance gains within neural networks without incurring the prohibitive costs associated within increasing the encoder model size via scaling techniques. For instance, in speech recognition tasks where the encoder includes an audio encoder and the output result is a transcription of an utterance characterized by the sequence of input tokens, the interleaving of the special think tokens can yield word error rate (WER) reductions comparable to those of models twice their size. As such, by circumventing the conventional size-performance tradeoff, the techniques of increasing computation per input token within an encoder disclosed herein aim to advance the capabilities of neural network models while ensuring their practicality and sustainability. The techniques of interleaving the special think tokens within the sequence of input tokens can be applied during both training and inference of the underlying model to ensure model robustness.

[0031] FIG. **1** is an example of a system **100** operating in a speech environment **101**. In the speech environment **101**, a user's **104** manner of interacting with a computing device, such as a user device **10**, may be through voice input. The user device **10** (also referred to generally as a device **10**) is configured to capture sounds (e.g., streaming audio data) from one or more users **104** within the speech environment **100**. Here, the streaming audio data may refer to a spoken utterance **106** by the user **104** that functions as an audible query, a command for the device **10**, or an audible communication captured by the device **10**. Speech-enabled systems of the device **10** may field the query or the command by answering the query and/or causing the command to be performed/fulfilled by one or more downstream applications.

[0032] The user device **10** may correspond to any computing device associated with a user **104** and capable of receiving audio data. Some examples of user devices **10** include, but are not limited to, mobile devices (e.g., mobile phones, tablets, laptops, etc.), computers, wearable devices (e.g., smart watches), smart appliances, internet of things (IoT) devices, vehicle infotainment systems, smart displays, smart speakers, etc. The user device **10** includes data processing hardware **12** and memory hardware **14** in communication with the data processing hardware **12** and storing instructions, that when executed by the data processing hardware **12**, causes the data processing hardware **12** to perform one or more operations. The user device **10** further includes an audio system **16** with an audio capture device (e.g., microphone) **16**, **16***a* for capturing and converting spoken utterances **106** within the speech environment **100** into electrical signals and a speech output device (e.g., a speaker) **16**, **16***b* for communicating an audible audio signal (e.g., as output audio data from the device **10**). While the user device **10** implements a single audio capture device **16***a* in the example shown, the user device **10** may implement an array of audio capture devices **16***a* without departing from the scope of the present disclosure, whereby one or more capture devices **16***a* in the array may not physically reside on the user device **10**, but be in communication with the audio system **16**.

[0033] In the speech environment **100**, an automated speech recognition (ASR) system **118** includes an ASR model **200** (such as an ASR model having a recurrent neural network-transducer (RNN-T model architecture or other transducer model/multi-pass model) that resides on the user device **10** of the user **104** and/or on a remote computing device **60** (e.g., one or more remote servers of a distributed system executing in a cloud-computing environment) in communication with the user device **10** via a network **40**. The remote computing device **60** is equipped with data processing hardware **62** and memory hardware **64**. The user device **10** and/or the remote computing device **60** also includes an audio subsystem **108** configured to receive the utterance **106** spoken by the user **104** and captured by the audio capture device **16***a*, and convert the utterance **106** into a corresponding digital format associated with input speech tokens (e.g., acoustic frames) **110** capable of being processed by the ASR system **118**. In the example shown, the user speaks a respective utterance **106** and the audio subsystem **108** converts the utterance **106** into corresponding speech tokens (e.g., acoustic frames) **110** for input to the ASR system **118**. Described in further detail below with reference to FIGS. **3** and **4**, the audio subsystem **108** further implements a contemplative mechanism **80** configured to insert a sequence of special think tokens **112** into the sequence of input speech tokens **110** such that the inserted special think tokens **112** are interleaved with the sequence of input speech tokens **110** prior to processing by an encoder **210** (FIGS. **2-4**) of the model **200** in order to increase per token computation within the encoder **210**. Accordingly, the model **200** receives, as input, the sequence of input speech tokens **110** corresponding to the utterance **106** interleaved with the sequence of special think tokens, and generates/predicts, as output, a corresponding transcription **120** (e.g., speech recognition result/hypothesis) of the utterance **106**. Notably, while the example of FIG. **1** depicts a streaming speech recognition example where the spoken utterance **106** is captured in streaming audio and processed in real-time, the ASR system **118** may similarly perform non-streaming speech recognition on the utterance after the user **104** finishes speaking the utterance **106** without departing from the scope of the present disclosure. Notably, in non-streaming applications, the utterance **106** may pertain to a pre-recorded utterance **106** input to the audio subsystem **108** as an audio file.

[0034] The user device **10** and/or the remote computing device **60** also executes a user interface generator **107** configured to present a representation of the transcription **120** of the utterance **106** to the user **104** of the user device **10**. As described in greater detail below, the user interface generator **107** may display the speech recognition results **120** in a streaming fashion. In some configurations, the transcription **120** output from the ASR system **118** is processed, e.g., by a natural language understanding (NLU) module executing on the user device **10** or the remote computing device **60**, to execute a user command/query specified by the utterance **106**. Additionally or alternatively, a text-to-speech system (not shown) (e.g., executing on any combination of the user device **10** or the remote computing device **60**) may convert the transcription into synthesized speech for audible output by the user device **10** and/or another device.

[0035] In the example shown, the user **104** interacts with a program or application **50** (e.g., the digital assistant application **50**) of the user device **10** that uses the ASR system **118**. For instance, FIG. **1** depicts the user **104** communicating with the digital assistant application **50** and the digital assistant application **50** displaying a digital assistant interface **18** on a screen of the user device **10** to depict a conversation between the user **104** and the digital assistant application **50**. In this example, the user **104** asks the digital assistant application **50**, "What time is the concert tonight?" This question from the user **104** is a spoken utterance **106** captured by the audio capture device **16***a* and processed by the audio systems **16** of the user device **10**. In this example, the audio subsystem **108** receives the spoken utterance **106** and converts it into acoustic frames **110** for input to the ASR system **118**.

[0036] Referring to FIG. **2**, an example frame alignment-based transducer model **200** includes a Recurrent Neural Network-Transducer (RNN-T) model architecture which adheres to latency constrains associated with interactive applications. The use of the RNN-T model architecture is exemplary, and the frame alignment-based transducer model **200** may include other architectures such as transformer-transducer and conformer-transducer model architectures among others. The RNN-T model **200** provides a small computational footprint and utilizes less memory requirements than conventional ASR architectures, making the RNN-T model architecture suitable for performing speech recognition entirely on the user device **102** (e.g., no communication with a remote server is required). The RNN-T model **200** includes an encoder network (i.e., audio encoder) **210**, a prediction network **220**, and a joint network **230**. The audio encoder **210**, which is roughly analogous to an acoustic model (AM) in a traditional ASR system, includes a stack of self-attention layers (e.g., Conformer or Transformer layers) or a recurrent network of stacked Long Short-Term Memory (LSTM) layers. For instance, the audio encoder **210** reads a sequence of d-dimensional feature vectors (e.g., speech tokens **110** (FIG. **1**)) $x=(x_1, x_2, \ldots, x_T)$, where $x_t \in \mathbb{R}^d$, interleaved with a sequence of a sequence of special tokens **112** (i.e., by the contemplation mechanism **80** (FIG. **1**) and produces at each output step a higher-order feature representation. This higher-order feature representation is denoted as $h_1^{enc}, \ldots, h_T^{enc}$ and may be interchangeably referred to as an 'encoder feature' **212**.

[0037] Similarly, the prediction network **220** is also an LSTM network, which, like a language model (LM), processes the sequence of non-blank symbols output by a final Softmax layer **240** so far, $y_0, \ldots, y_{u_i-1}$, into a dense representation p_{u_i}. Finally, with the RNN-T model architecture, the representations produced by the encoder and prediction/decoder networks **210**, **220** are combined by the joint network **230**. While not shown, the sequence of encoder features **212** output by the encoder may be down-sampled prior to processing by the joint network **230** so that a number of the encoder-features after down-sampling is equal to a number of the speech tokens **110** in the sequence of input speech tokens **110** The prediction network **220** may be replaced by an embedding look-up table to improve latency by outputting looked-up sparse embeddings in lieu of processing dense representations.

[0038] The joint network then predicts $P(y_i|x_{t_i}, y_0, \ldots, y_{u_{i-1}})$, which is a distribution over the next output symbol. Stated differently, the joint network **230** generates, at each output step (e.g., time step), a probability distribution over possible speech recognition hypotheses Here, the "possible speech recognition hypotheses" correspond to a set of output labels each representing a symbol/character in a specified natural language. For example, when the natural language is English, the set of output labels may include twenty-seven (27) symbols, e.g., one label for each of the 26-letters in the English alphabet and one label designating a space. Accordingly, the joint network **230** may output a set of values indicative of the likelihood of occurrence of each of a predetermined set of output labels. This set of values can be a vector and can indicate a probability distribution over the set of output labels. In some cases, the output labels are graphemes (e.g., individual characters, and potentially punctuation and other symbols), but the set of output labels is not so limited. For example, the set of output labels can include wordpieces, phonemes, and/or entire words, in addition to or instead of graphemes. The output distribution of the joint network **230** can include a posterior probability value for each of the different output labels. Thus, if there are 100 different output labels representing different graphemes or other symbols, the output y_i of the joint network **230** can include 100 different probability values, one for each output label. The probability distribution can then be used to select and assign scores to candidate orthographic elements (e.g., graphemes, wordpieces, and/or words) in a beam search process (e.g., by the Softmax layer **240**) for determining the transcription **120**.

[0039] The Softmax layer **240** may employ any technique to select the output label/symbol with the highest probability in the distribution as the next output symbol predicted by the RNN-T model **200** at the corresponding output step. In this manner, the RNN-T model **200** does not make a conditional independence assumption, rather the prediction of each symbol is conditioned not only on the acoustics but also on the sequence of labels output so far. The RNN-T model **200** does assume an output symbol is independent of future acoustic frames **110**, which allows the RNN-T model to be employed in a streaming fashion.

[0040] In some examples, the encoder network (i.e., audio encoder) **210** of the RNN-T model **200** includes a stack of self-attention layers/blocks, such as conformer layers/blocks. In some examples, the number of self-attention layers/blocks in the audio encoder is equal to 24. For instance, the audio encoder **210** may include 150 million parameters having 24 conformer layers with 512 hidden dimensions, 300 million parameters having 24 conformer

layers with 768 hidden dimensions, or 600 million parameters having 24 conformer layers with 1,024 hidden dimensions. In some examples, the relative attention in each self-attention layer is equal to 16 attention heads. Here, each conformer block includes a series of multi-headed self attention, depth wise convolution and feed-forward layers. The stack of self-attention layers/blocks may include transformer layers/blocks in other examples. The prediction network **220** may have two 2,048-dimensional LSTM layers, each of which is also followed by 640-dimensional projection layer. Alternatively, the prediction network **220** may include a stack of transformer or conformer blocks, or an embedding look-up table in lieu of LSTM layers. Finally, the joint network **230** may also have 640 hidden units. The Softmax layer **240** may be composed of a unified word piece or grapheme set that is generated using all unique word pieces or graphemes in a plurality of training data sets. The prediction network **220**, the joint network **230**, and the Softmax layer **240** may collectively form an RNN-T decoder **250** of the RNN-T model **200**.

[0041] FIG. **3** illustrates an example training process **300** for training the audio encoder **210** of the ASR model **200** (FIG. **2**). Notably, the audio encoder **210** may be initially pre-trained and the training process **300** trains/fine-tunes the audio encoder **210** from the pre-trained checkpoint. For instance, the audio encoder **210** may be pre-trained on any combination of text-only utterances not paired any spoken representations of the utterances, un-transcribed speech utterances not paired with any corresponding transcriptions of the utterances, and transcribed speech utterances each including a speech representation of an utterance paired with a corresponding transcription of the utterance. An example pre-training process for pre-training the audio encoder **210** is disclosed in commonly assigned U.S. patent application Ser. No. 18/585,168, filed on Feb. 23, 2024, the contents of which are incorporated by reference its entirety.

[0042] The training process **300** trains the audio encoder **210** based on supervised losses (L_{aux}) **344** derived from a set of transcribed speech utterances (X_{sup}) **304**, **304***a-n*. Each transcribed speech utterance **304** includes a corresponding transcription **302** paired with a corresponding training input token sequence **310** characterizing a speech representation of the corresponding transcribed speech utterance **304**. Each corresponding training input token sequence **310** includes a corresponding sequence of training input tokens **310**, **310***a-n*. Here, the training input tokens **310** may include training input speech tokens **310** each representing a respective acoustic frame.

[0043] For each training input token sequence **310** of each transcribed speech utterance **304**, the contemplation mechanism **80** inserts a corresponding sequence of training think tokens **312** into the training input token sequence **310** such that the corresponding sequence of training think tokens **312** are interleaved with the corresponding sequence of training input tokens **310***a-n*. For each corresponding training input token **310**, the contemplation mechanism **80** may insert the corresponding sequence of training think tokens **312** by appending or prepending a fixed number of the training think tokens **312** to the corresponding training input token **310**. In the example shown, the contemplation mechanism **80** appends two think tokens **312** to each training input token **310**. Notably, the contemplation mechanism **80** may append only one think token or more than three think tokens **312** to each training input token **310**, or instead, the contemplation mechanism **80** may prepend one, two, or more than two think tokens **312** to each training input token **310** without departing from the scope of the present disclosure.

[0044] In some implementations, the training process **300** gradually increases a ratio of the corresponding sequence of training think tokens inserted into each training input token sequence **310**. Here, an annealing technique gradually increases the ratio of the corresponding sequence of think tokens **312** according to an annealing schedule when the training process **300** is fine-tuning the audio encoder **210** from the pre-trained checkpoint. The annealing schedule may lead to faster convergence and reduced error rates of the model **200** during the training process **300** compared to using a constant think token ratio where the fixed number of think tokens appended or prepended to each training input token **310** remains static during the entire training process **300**

[0045] Next, the training process **300** processes, using the audio encoder **210**, the training input token sequence **310** interleaved with the corresponding sequence of training think tokens **312** to generate a sequence of training encoder features h **212**. The audio encoder **210** may generate a training encoder feature **210** from a respective speech token **310** or think token **312** at each corresponding time step. Notably, a number of the training encoder features **210** is equal to a sum of a number of the training input tokens **310***a-n* in the corresponding sequence of training input tokens **310** and a number of the think tokens **312** in the sequence of special tokens **312**. Thus, as the number of training encoder features **210** is greater than the number of training input tokens in the corresponding sequence of training input tokens **310**, a down sample mechanism **85** may down-sample the sequence of training encoder features h **212** output by the encoder **210** to obtain a down-sampled encoder feature sequence **212**D. Here, the down-sampled encoder feature sequence **212**D includes a number of encoder features **212** that is equal to the number of training input tokens **310** in the sequence of training input tokens **310**. Thereafter, the decoder **250** processes the down-sampled encoder feature sequence **212**D to generate an output result **394**. Specifically, the decoder **250** receives, as input, each encoder feature **212** from the down-sampled encoder feature sequence **212**D and generates, as output, a corresponding output result corresponding to a probability distribution over possible speech recognition hypotheses **394** for the corresponding transcribed speech utterance **304** at the corresponding time step.

[0046] Notably, the decoder **250** may include a phoneme decoder configured to decode a sequence of phonemes or a word piece decoder configured to decode a sequence of word pieces. The decoder **250** could also include a grapheme decoder configured to decode a sequence of graphemes or other sub-word units. While the decoder **250** may include the RNN-T decoder architecture depicted in FIG. **2**, the decoder may also include other types of decoders such as a Connectionist Temporal Classification (CTC) decoder, a Listen-Attend-Spell (LAS) decoder, or a large language model (LLM) decoder.

[0047] In some examples, the probability distribution over possible speech recognition hypotheses **394** includes the one of possible phoneme labels, the possible word piece labels, or the possible grapheme labels. Thereafter, a supervised loss module **340** may determine a speech loss term **344** based on the probability distribution over possible speech

recognition hypotheses **394** and the corresponding transcription **302** paired with the corresponding training input token sequence **310** characterizing thea speech representation of the corresponding transcribed speech utterance **304**. Here, the corresponding transcription **302** serves as a ground-truth transcription and may include a sequence of target phonemes, target word pieces, and/or target graphemes. When the ASR model **200** includes the CTC decoder **250**, the speech loss term **344** includes a CTC loss. When the ASR model **200** includes the RNN-T model **200** that includes the RNN-T decoder **250** of FIG. **2**, the speech loss term **344** includes an RNN-T loss. The training process **300** trains the model **200** based on the supervised speech loss terms **344** by updating parameters of the audio encoder **210** and/or the decoder **250** using the supervised speech loss terms **344**.

[0048] FIG. **4** shows a schematic view **400** of the model **200** performing speech recognition on the utterance **106** spoken by the user **104** and captured by the audio capture device **16***a* of FIG. **1** after the model **200** is trained by the training process **300** of FIG. **3**. The audio subsystem **108** is configured to receive the utterance **106** spoken by the user **104** and captured by the audio capture device **16***a*, and convert the utterance **106** into a corresponding digital format associated with the input speech tokens **110**. In the example shown, the user speaks a respective utterance **106** and the audio subsystem **108** converts the utterance **106** into a sequence of speech tokens **110**, **110***a-n* for input to the ASR model **200**. Next, the contemplation mechanism **80** inserts a sequence of special think tokens **112** into the sequence of speech tokens **110** such that the special think tokens **112** are interleaved with the sequence of speech tokens **110**. For each corresponding speech token **110**, the contemplation mechanism **80** may insert the special think tokens **112** by appending or prepending a fixed number of the special think tokens **112** to the corresponding speech token **110**. In the example shown, the contemplation mechanism **80** prepends one think tokens **112** to each speech token **110**. Notably, the contemplation mechanism **80** may prepend two or more think tokens **112** to each speech token **110**, or instead, the contemplation mechanism **80** may append one or more think tokens **312** to each speech input token **310** without departing from the scope of the present disclosure.

[0049] Appending think tokens **112** allows for processing by the encoder **210** to rely on hidden vectors generated for the corresponding speech token. Conversely, prepending think tokens **112** offers the encoder **210** visibility into additional computational outcomes before directly processing the corresponding speech token **110**, which may provide a potentially beneficial relationship. As the audio encoder **210** may include limited left and right context windows, the left and right context windows may be increased proportionally to the number of think tokens **110** is appending or prepending to each corresponding speech token **110**. By increasing the left and right context windows proportional to the number of think tokens **110**, parity may be maintained in terms of the original speech tokens **110** visible to the audio encoder **210**.

[0050] With continued reference to FIG. **4**, the audio encoder **210** processes the sequence of speech tokens **110** interleaved with the special think tokens **112** to generate a sequence of encoder features h **212**. The audio encoder **210** may generate each encoder feature **210** from a respective speech token **110** or think token **112** at each corresponding time step Notably, a number of the encoder features **210** is equal to a sum of a number of the speech tokens **110***a-n* in the corresponding sequence of speech tokens **110** and a number of the think tokens **112** in the sequence of special tokens **312**. Thus, as the number of encoder features **210** is greater than the number of speech tokens **112** in the sequence of speech tokens **110** characterizing the spoken utterance **106**, a down sample mechanism **85** may down-sample the sequence of encoder features h **212** output by the encoder **210** to obtain a down-sampled encoder feature sequence **212**D. Here, the down-sampled encoder feature sequence **212**D includes a number of encoder features **212** that is equal to the number of speech tokens **110** in the sequence of speech tokens **110**. Thereafter, the decoder **250** processes the down-sampled encoder feature sequence **212**D to generate an output result **194**. Specifically, the decoder **250** receives, as input, each encoder feature **212** from the down-sampled encoder feature sequence **212**D and generates, as output, a corresponding output result **194** corresponding to a probability distribution over possible speech recognition hypotheses for the spoken utterance **106** at the corresponding time step. Ultimately, the probability distributions over possible speech recognition hypotheses generated by processing all of the down-sampled encoder features **212**D form a transcription **120** of the utterance **106**. For instance, a beam search may select the transcription **120** from a lattice of candidate transcriptions based on the probability distributions over possible speech recognition hypotheses. Notably, the decoder **250** may include a phoneme decoder configured to decode a sequence of phonemes or a word piece decoder configured to decode a sequence of word pieces. The decoder **250** could also include a grapheme decoder configured to decode a sequence of graphemes or other sub-word units. While the decoder **250** may include the RNN-T decoder architecture depicted in FIG. **2**, the decoder may also include other types of decoders such as a Connectionist Temporal Classification (CTC) decoder, a Listen-Attend-Spell (LAS) decoder, or a large language model (LLM) decoder. Notably, contemplation mechanism **80** may insert the special think tokens **112** as the speech tokens **110** are received from the audio subsystem **108** in a streaming fashion to support a streaming speech encoder **210**. The speech encoder **210** may also operate in a non-streaming mode by processing the speech tokens **110** interleaved with the special think tokens **112**. Accordingly, the contemplation mechanism **80** for interleaving the special think tokens **112** can be applied equivalently by both causal (streaming) encoders and non-causal (non-streaming) encoders.

[0051] The down-sampling performed by the down sample mechanism **85** may be optional. When down sampling is not performed on the original encoder features **210** output from the encoder **210**, the decoder **250** simply processes the longer sequence of encoder features **210** which may amplify benefits of enhanced feature representation.

[0052] While the model **200** of FIGS. **2-4** is depicted as an ASR model, the model **200** may include other types of speech models without departing from the scope of the present disclosure. For instance, the model **200** may instead include a speech translation model that includes a decoder configured to process the down-sampled encoder features **212**D to generate a translation of the utterance **106** in a different language than a language of the utterance **106** spoken by the user **104**. In another example, the model **200** may include a speech to speech model that includes a

decoder configured to process the down-sampled encoder features **212**D to generate a synthesized speech representation of the utterance **106** in a different voice than a voice of the user **105** that spoke the utterance **106** and/or in a different language than a language of the utterance **106** of the utterance **106** spoken by the user **104**. In other examples, the model **200** may include a keyword detection model where a decoder processes the down-sampled encoder features **212**D to determine whether a keyword is present in the utterance **106**. In even further examples, the model **200** may include a speaker identification model where the down-sampled encoder features **212** are processed to determine a speaker vector characterizing a voice of the user **104** that spoke the utterance **106**. Other speech applications include speech classification models that determine a single prediction per speech sequence such as emotion recognition, language identification, speaker classification, demographic classification, speaker diarization, speaker change detection, disfluency annotation, and speaker state classification to name a few. The present disclosure is not limited to speech, whereby the technique of interleaving a sequence of input tokens (e.g., text tokens, image tokens, etc.) with think tokens can be utilized for other applications such as text-to-text applications or image recognition.

[0053] In some implementations, the think tokens **112**, **312** inserted by the contemplation mechanism **80** during inference (FIGS. **1** and **4**) and training (FIG. **3**) are handcrafted think tokens Handcrafted tokens may include think tokens that are predetermined by a user/developer such as zero-tensor think tokens, a mean-feature value of the speech tokens, a fixed value learned from an external model and repeated for each think token **112**, a position-derived embedding, or any combination.

[0054] In some additional implementations, the think tokens **112**, **312** include duplications of the input tokens **110**, **310**. Here, the input tokens **110**, **310** characterizing input speech representations are repeated for k frames where k is equal to the number of think tokens **112**, **312**. Thus, an input token sequence of A-B-C would be presented as input to the encoder **21** as AAA-BBB-CCC if k is equal to two (2).

[0055] In other implementations, the think tokens **112**, **312** include learned parameters. Here, the think tokens **112**, **312** are treated as a trainable variable to be shared across all think tokens, but optimized along with the objective function of the encoder **210**. In these implementations, a standard variable initialization (e.g., structured random initialization) may be applied to obtain initial variables for the think tokens and then updating is applied across all presentations of each think token. As an alternative to updating the variables directly, the think tokens include a fixed initial variable and an adapter may update the fixed initial variable. The adapter could include a residual adapter, low-rank adaptation (LoRA), or residual bottleneck. In other examples, a large language model (LLM) is leveraged to predict the values for the learned parameters representing the think tokens **112**, **312**.

[0056] In some implementations, the presentation of think tokens **110**, **310** are alternatively modified by a positional embedding. The position embedding may be additive or concatenated to the think tokens **110**, **310**. The positional embedding can follow the original sequence length and can be extended as the original sequence (e.g., 1, 1, 1, 2, 2, 2, 3, 3, 3) or applied to the resulting extended sequence (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9).

[0057] FIG. **5** shows an example plot **500** evaluating speech recognition performance of three ASR models without (Original) implementing the contemplative mechanism **80** with implementing (Contemplative) the contemplative mechanism **80**. The x-axis denotes model size in millions (M) and the y-axis denotes word error rate (WER) The three ASR models implement the audio encoder **210** having 24 Conformer layers with varying sizes of 150 million (M) parameters, 300 million (M) parameters, and 600 million (M) parameters, and the RNN-T decoder **250** of FIG. **2**. For the ASR models implementing the contemplative mechanism **80**, the contemplative mechanism **80** prepends a single learned parameter think token to each original speech token. Additionally, the context window is doubled and the encoder features **212** output by the encoder are down-sampled to accommodate the expanded sequence length. Notably, the insertion of the single think token per each speech token yields a consistent WER reduction across all three model sizes, thereby enabling smaller models to achieve accuracy levels comparable to those of models twice their size. For instance, the WER of the **150**M parameter model utilizing the contemplative mechanism **80** closely matches the WER of the **300**M parameter model not utilizing the contemplative mechanism **80**. Moreover, the WER of the **300**M parameter model utilizing the contemplative mechanism **80** outperforms the **600**M model not implementing the contemplative mechanism **80**.

[0058] FIG. **6** is a flowchart of an exemplary arrangement of operations for a method **600** of increasing computation per speech token within an encoder while maintaining an original parameter count of the encoder. The method **600** may execute on data processing hardware **710** (FIG. **7**) based on instructions stored on memory hardware **720** (FIG. **7**). The data processing hardware **710** may include the data processing hardware **62** of the remote computing system **60** or the data processing hardware **12** of the user device **10**. The memory hardware **720** may include the memory hardware **64** of the remote computing system **60** or the memory hardware **14** of the user device **10**.

[0059] At operation **602**, the method **600** includes receiving a sequence of input tokens **110**. The input tokens may include speech tokens characterizing a sequence of acoustic frames characterizing a spoken utterance **106**. At operation **604**, the method **600** includes inserting a sequence of special think tokens **112** into the sequence of input tokens **110**. The inserted special think tokens **112** are interleaved with the sequence of input tokens **110**. For instance, a fixed number of special think tokens **112** may be appended or prepended to each corresponding input token **110** in the sequence of input tokens **110**.

[0060] At operation **606**, the method **600** includes processing, using an encoder **210**, the sequence of input tokens **110** interleaved with the sequence of special think tokens **112** to generate a sequence of encoder features **212**. At operation **608**, the method includes processing the sequence of encoder features **210** to generate an output result. The output result may include a transcription of the spoken utterance. The output result could also include a synthetic speech translation of the spoken utterance.

[0061] FIG. **7** is a schematic view of an example computing device **700** that may be used to implement the systems and methods described in this document. The computing device **700** is intended to represent various forms of digital computers, such as laptops, desktops, workstations, personal

digital assistants, servers, blade servers, mainframes, and other appropriate computers. The components shown here, their connections and relationships, and their functions, are meant to be exemplary only, and are not meant to limit implementations of the inventions described and/or claimed in this document.

[0062] The computing device **700** includes a processor **710**, memory **720**, a storage device **730**, a high-speed interface/controller **740** connecting to the memory **720** and high-speed expansion ports **750**, and a low speed interface/controller **760** connecting to a low speed bus **770** and a storage device **730**. Each of the components **710**, **720**, **730**, **740**, **750**, and **760**, are interconnected using various busses, and may be mounted on a common motherboard or in other manners as appropriate. The processor **710** can process instructions for execution within the computing device **700**, including instructions stored in the memory **720** or on the storage device **730** to display graphical information for a graphical user interface (GUI) on an external input/output device, such as display **780** coupled to high speed interface **740**. In other implementations, multiple processors and/or multiple buses may be used, as appropriate, along with multiple memories and types of memory. Also, multiple computing devices **700** may be connected, with each device providing portions of the necessary operations (e.g., as a server bank, a group of blade servers, or a multi-processor system).

[0063] The memory **720** stores information non-transitorily within the computing device **700**. The memory **720** may be a computer-readable medium, a volatile memory unit(s), or non-volatile memory unit(s). The non-transitory memory **720** may be physical devices used to store programs (e.g., sequences of instructions) or data (e.g., program state information) on a temporary or permanent basis for use by the computing device **700**. Examples of non-volatile memory include, but are not limited to, flash memory and read-only memory (ROM)/programmable read-only memory (PROM)/erasable programmable read-only memory (EPROM)/electronically erasable programmable read-only memory (EEPROM) (e.g., typically used for firmware, such as boot programs). Examples of volatile memory include, but are not limited to, random access memory (RAM), dynamic random access memory (DRAM), static random access memory (SRAM), phase change memory (PCM) as well as disks or tapes.

[0064] The storage device **730** is capable of providing mass storage for the computing device **700**. In some implementations, the storage device **730** is a computer-readable medium. In various different implementations, the storage device **730** may be a floppy disk device, a hard disk device, an optical disk device, or a tape device, a flash memory or other similar solid state memory device, or an array of devices, including devices in a storage area network or other configurations. In additional implementations, a computer program product is tangibly embodied in an information carrier. The computer program product contains instructions that, when executed, perform one or more methods, such as those described above. The information carrier is a computer- or machine-readable medium, such as the memory **720**, the storage device **730**, or memory on processor **710**.

[0065] The high speed controller **740** manages bandwidth-intensive operations for the computing device **700**, while the low speed controller **760** manages lower bandwidth-intensive operations. Such allocation of duties is exemplary only. In some implementations, the high-speed controller **740** is coupled to the memory **720**, the display **780** (e.g., through a graphics processor or accelerator), and to the high-speed expansion ports **750**, which may accept various expansion cards (not shown). In some implementations, the low-speed controller **760** is coupled to the storage device **730** and a low-speed expansion port **790**. The low-speed expansion port **790**, which may include various communication ports (e.g., USB, Bluetooth, Ethernet, wireless Ethernet), may be coupled to one or more input/output devices, such as a keyboard, a pointing device, a scanner, or a networking device such as a switch or router, e.g., through a network adapter.

[0066] The computing device **700** may be implemented in a number of different forms, as shown in the figure. For example, it may be implemented as a standard server **700***a* or multiple times in a group of such servers **700***a*, as a laptop computer **700***b*, or as part of a rack server system **700***c*.

[0067] Various implementations of the systems and techniques described herein can be realized in digital electronic and/or optical circuitry, integrated circuitry, specially designed ASICs (application specific integrated circuits), computer hardware, firmware, software, and/or combinations thereof. These various implementations can include implementation in one or more computer programs that are executable and/or interpretable on a programmable system including at least one programmable processor, which may be special or general purpose, coupled to receive data and instructions from, and to transmit data and instructions to, a storage system, at least one input device, and at least one output device.

[0068] A software application (i.e., a software resource) may refer to computer software that causes a computing device to perform a task. In some examples, a software application may be referred to as an "application," an "app," or a "program." Example applications include, but are not limited to, system diagnostic applications, system management applications, system maintenance applications, word processing applications, spreadsheet applications, messaging applications, media streaming applications, social networking applications, and gaming applications.

[0069] These computer programs (also known as programs, software, software applications or code) include machine instructions for a programmable processor, and can be implemented in a high-level procedural and/or object-oriented programming language, and/or in assembly/machine language. As used herein, the terms "machine-readable medium" and "computer-readable medium" refer to any computer program product, non-transitory computer readable medium, apparatus and/or device (e.g., magnetic discs, optical disks, memory, Programmable Logic Devices (PLDs)) used to provide machine instructions and/or data to a programmable processor, including a machine-readable medium that receives machine instructions as a machine-readable signal. The term "machine-readable signal" refers to any signal used to provide machine instructions and/or data to a programmable processor.

[0070] The processes and logic flows described in this specification can be performed by one or more programmable processors, also referred to as data processing hardware, executing one or more computer programs to perform functions by operating on input data and generating output. The processes and logic flows can also be performed by special purpose logic circuitry, e.g., an FPGA (field pro-

grammable gate array) or an ASIC (application specific integrated circuit). Processors suitable for the execution of a computer program include, by way of example, both general and special purpose microprocessors, and any one or more processors of any kind of digital computer. Generally, a processor will receive instructions and data from a read only memory or a random access memory or both. The essential elements of a computer are a processor for performing instructions and one or more memory devices for storing instructions and data. Generally, a computer will also include, or be operatively coupled to receive data from or transfer data to, or both, one or more mass storage devices for storing data, e.g., magnetic, magneto optical disks, or optical disks. However, a computer need not have such devices. Computer readable media suitable for storing computer program instructions and data include all forms of non-volatile memory, media and memory devices, including by way of example semiconductor memory devices, e.g., EPROM, EEPROM, and flash memory devices; magnetic disks, e.g., internal hard disks or removable disks; magneto optical disks; and CD ROM and DVD-ROM disks. The processor and the memory can be supplemented by, or incorporated in, special purpose logic circuitry.

[0071] To provide for interaction with a user, one or more aspects of the disclosure can be implemented on a computer having a display device, e.g., a CRT (cathode ray tube), LCD (liquid crystal display) monitor, or touch screen for displaying information to the user and optionally a keyboard and a pointing device, e.g., a mouse or a trackball, by which the user can provide input to the computer. Other kinds of devices can be used to provide interaction with a user as well; for example, feedback provided to the user can be any form of sensory feedback, e.g., visual feedback, auditory feedback, or tactile feedback; and input from the user can be received in any form, including acoustic, speech, or tactile input. In addition, a computer can interact with a user by sending documents to and receiving documents from a device that is used by the user; for example, by sending web pages to a web browser on a user's client device in response to requests received from the web browser.

[0072] A number of implementations have been described. Nevertheless, it will be understood that various modifications may be made without departing from the spirit and scope of the disclosure. Accordingly, other implementations are within the scope of the following claims.

What is claimed is:

1. A computer-implemented method executed on data processing hardware that causes the data processing hardware to perform operations comprising:

receiving a sequence of input tokens;

inserting a sequence of special think tokens into the sequence of input tokens, wherein the inserted special think tokens are interleaved with the sequence of input tokens;

processing, using an encoder, the sequence of input tokens interleaved with the sequence of special think tokens to generate a sequence of encoder features; and

processing the sequence of encoder features to generate an output result.

2. The computer-implemented method of claim **1**, wherein inserting the sequence of special tokens comprises, for each corresponding input token in the sequence of input tokens, prepending a fixed number of the special think tokens to the corresponding input token.

3. The computer-implemented method of claim **1**, wherein inserting the sequence of special tokens comprises, for each corresponding input token in the sequence of input tokens, prepending a fixed number of the special think tokens to the corresponding input token.

4. The computer-implemented method of claim **1**, wherein the sequence of input tokens comprises a sequence of speech tokens characterizing a spoken utterance and the encoder comprises a speech encoder.

5. The computer-implemented method of claim **4**, wherein the output result comprises a transcription of the spoken utterance.

6. The computer-implemented method of claim **4**, wherein the output result comprises a synthetic speech translation of the spoken utterance.

7. The computer-implemented method of claim **1**, wherein the special think tokens comprise handcrafted think tokens.

8. The computer-implemented method of claim **1**, wherein:

the sequence of special think tokens comprise duplications of the sequence of input tokens; and

inserting the sequence of special tokens comprises, for each corresponding input token in the sequence of input tokens:

duplicating the corresponding input token; and

appending or prepending one or more duplications of the corresponding input token to the corresponding input token.

9. The computer-implemented method of claim **1**, wherein the special think tokens comprise learnable parameters.

10. The computer-implemented method of claim **1**, wherein:

a number of the encoder features in the sequence of encoder features is equal to a sum of a number of the input tokens in the sequence of input tokens and a number of the special think tokens in the sequence of special think tokens; and

processing the sequence of encoder features to generate the output result comprises:

down-sampling the sequence of encoder features to obtain a down-sampled encoder feature sequence, wherein a number of the encoder features in the down-sampled encoder feature sequence is equal to the number of the input tokens in the sequence of input tokens; and

processing, using a decoder, the down-sampled encoder feature sequence to generate the output result.

11. The computer-implemented method of claim **1**, wherein processing the sequence of encoder features to generate the output result comprises processing, using a decoder, the sequence of encoder features to generate the output result.

12. The computer-implemented method of claim **11**, wherein a number of the encoder features in the sequence of encoder features is greater than a number of the input tokens in the sequence of input tokens.

13. The computer-implemented method of claim **1**, wherein a model implementing the encoder is trained by a training process that comprises:

obtaining a set of set of training input token sequences, each training input token sequence comprising a corresponding sequence of training input tokens

for each training input token sequence:
inserting a corresponding sequence of training think tokens into the training input token sequence such that corresponding sequence of training think tokens are interleaved with the corresponding sequence of training input tokens; and
processing, using the encoder, the training input token sequence interleaved with the corresponding sequence of training think tokens to generate a sequence of training encoder features; and
training the model based on the sequence of training encoder features generated for each training input token sequence.

14. The computer-implemented method of claim **13**, wherein a ratio of the corresponding sequence of training think tokens inserted into each training input token sequence gradually increases during the training process.

15. The computer-implemented method of claim **13**, wherein the model comprises a recurrent neural network-transducer (RNN-T) architecture comprising the encoder, a prediction network, and a joint network.

16. A system comprising:
data processing hardware; and
memory hardware in communication with the data processing hardware and storing instructions that when executed on the data processing hardware cause the data processing hardware to perform operations comprising:
receiving a sequence of input tokens;
inserting a sequence of special think tokens into the sequence of input tokens, wherein the inserted special think tokens are interleaved with the sequence of input tokens;
processing, using an encoder, the sequence of input tokens interleaved with the sequence of special think tokens to generate a sequence of encoder features; and
processing the sequence of encoder features to generate an output result.

17. The system of claim **16**, wherein inserting the sequence of special tokens comprises, for each corresponding input token in the sequence of input tokens, prepending a fixed number of the special think tokens to the corresponding input token.

18. The system of claim **16**, wherein inserting the sequence of special tokens comprises, for each corresponding input token in the sequence of input tokens, prepending a fixed number of the special think tokens to the corresponding input token.

19. The system of claim **16**, wherein the sequence of input tokens comprises a sequence of speech tokens characterizing a spoken utterance and the encoder comprises a speech encoder.

20. The system of claim **19**, wherein the output result comprises a transcription of the spoken utterance.

21. The system of claim **19**, wherein the output result comprises a synthetic speech translation of the spoken utterance.

22. The system of claim **16**, wherein the special think tokens comprise handcrafted think tokens.

23. The system of claim **16**, wherein:
the sequence of special think tokens comprise duplications of the sequence of input tokens; and
inserting the sequence of special tokens comprises, for each corresponding input token in the sequence of input tokens:
duplicating the corresponding input token; and
appending or prepending one or more duplications of the corresponding input token to the corresponding input token.

24. The system of claim **16**, wherein the special think tokens comprise learnable parameters.

25. The system of claim **16**, wherein:
a number of the encoder features in the sequence of encoder features is equal to a sum of a number of the input tokens in the sequence of input tokens and a number of the special think tokens in the sequence of special think tokens; and
processing the sequence of encoder features to generate the output result comprises:
down-sampling the sequence of encoder features to obtain a down-sampled encoder feature sequence, wherein a number of the encoder features in the down-sampled encoder feature sequence is equal to the number of the input tokens in the sequence of input tokens; and
processing, using a decoder, the down-sampled encoder feature sequence to generate the output result.

26. The system of claim **16**, wherein processing the sequence of encoder features to generate the output result comprises processing, using a decoder, the sequence of encoder features to generate the output result.

27. The system of claim **26**, wherein a number of the encoder features in the sequence of encoder features is greater than a number of the input tokens in the sequence of input tokens.

28. The system of claim **16**, wherein a model implementing the encoder is trained by a training process that comprises:
obtaining a set of set of training input token sequences, each training input token sequence comprising a corresponding sequence of training input tokens
for each training input token sequence:
inserting a corresponding sequence of training think tokens into the training input token sequence such that corresponding sequence of training think tokens are interleaved with the corresponding sequence of training input tokens, and
processing, using the encoder, the training input token sequence interleaved with the corresponding sequence of training think tokens to generate a sequence of training encoder features; and
training the model based on the sequence of training encoder features generated for each training input token sequence.

29. The system of claim **28**, wherein a ratio of the corresponding sequence of training think tokens inserted into each training input token sequence gradually increases during the training process.

30. The system of claim **28**, wherein the model comprises a recurrent neural network-transducer (RNN-T) architecture comprising the encoder, a prediction network, and a joint network.

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